

JWST VS HUBBLE VS KEPLER VS TESS

All of these are primarily NASA space science missions. The Kepler mission is a retired space telescope launched in 2009, and it was designed to find what fraction of stars have earth-like planets near the habitable zone. Launched in 2018, TESS (Transiting Exoplanet Survey Satellite) practically expands on Kepler mission's findings, trying to find exoplanets orbiting stars that are closer to our solar system for further observation and study. Technically, JWST can do all of this but its field of scientific inquiry is a lot more fundamental – kind of like the Hubble Space Telescope it's supposed to replace shortly.

Think of it this way – the Hubble Space Telescope is the first non-terrestrial, space-based telescope launched and has been continuously operated since 1990. It's essentially a giant magnifying glass suspended in space to reduce earth-based radio interference and look at a wide variety of objects

in the universe. Hubble's findings have led to future missions like Kepler and TESS.

If you thought Hubble was good, JWST is at least a hundred times better than the Hubble Space Telescope. By viewing the universe at infrared wavelengths, the JWST aims to show us things never before seen by any other telescope – like the very first stars and galaxies formed within 100 million years of the Big Bang (which took place over 14 billion years ago). JWST isn't just a space telescope, it's a full-fledged observatory that's far away from earth's electromagnetic interference – ideal to study how galaxies, stars and planets formed in the early universe. That's why JWST is an infrared-based telescope, because infrared light allows us to see stars and planetary systems forming inside clouds of dust that aren't fully visible in ordinary light that we or our eyes can see.

Biden, during the official unveiling of the image. "If you held a grain of sand on the tip of your finger at arm's length, that is the part of the universe that you're seeing (in this picture)... just one little speck of the universe," NASA Administrator Bill Nelson said. "You're seeing galaxies that are shining around other galaxies whose light has been bent."

As sophisticated as the James Webb Space Telescope is, fundamentally it's aimed at doing space imaging largely in the infrared spectrum of light. This fact is confirmed by JWST's official Twitter handle, too. It says that Webb's First Deep Field image "isn't the farthest back we've observed" the universe. "Non-infrared missions like COBE & WMAP saw

the universe closer to the Big Bang (~380,000 years after), when there was only microwave background radiation, but no stars or galaxies. Webb sees a few 100 million years after the Big Bang," it highlighted in no uncertain terms.

WHAT'S NEXT FOR JWST

According to NASA's Bill Nelson, the James Webb Space Telescope is just getting warmed up, just getting started to do what the Hubble Space Telescope could never physically do – which is to look back at the very beginning of the universe immediately after the Big Bang.

This isn't all, as James Webb Space Telescope's nothing short of a full-fledged science lab in deep space. Its on-board cutting edge capabilities will help NASA researchers on ground here on Earth analyze not just static images of the early universe, but also the chemical composition of the atmosphere of planets beyond our solar system, and whether they can sustain life on them. Humanity's best telescope, the JWST is scheduled to operate for 10 years at the bare minimum, but knowing NASA's ingenuity don't be surprised if we continue to see it operating for several decades into the future – for the sake of unprecedented space science, we sure hope so! **d**

MORE IMAGES FROM JWST



Carina Nebula - Like a painting in space



Southern Ring Nebula - Some stars save the best for last



SMACS 0723 - Very far away... There is a collection of galaxies



Stephen's Quintet - Five galaxies in a single place